

Parsa Abbasian

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EDUCATION

York University

Toronto, ON

Bachelor of Engineering (B.Eng.) in Computer Engineering

Sep. 2025 – Apr. 2029

- **Academic Standing:** GPA: A (Top-tier academic performance)
- Expected specialization in Artificial Intelligence & Machine Learning.
- Relevant Coursework: Applied Calculus I & II, Discrete Mathematics, Linear Algebra, Programming for Engineers.

EXPERIENCE

Undergraduate Researcher — Space Geodesy Lab

Jan. 2026 – Present

York University — Research Fellowship

Toronto, ON

- Conducting data-driven research on extreme weather events (hurricanes) using satellite observations and climate model datasets.
- Engineering scientific Python workflows for automated data acquisition and multi-dimensional analysis.
- Developing AI-driven machine learning models to characterize hurricane evolution and assess infrastructure impact.
- Working under the supervision of **Dr. Spiros Pagiatakis** on environmental risk modeling and storm prediction reliability.

Vice President — MBH&Co.

Jan. 2026 – Present

B2B Subscription-Based Digital Business Directory Platform

Toronto, ON

- **Co-founded and architected** a **B2B subscription-based digital directory platform** enabling businesses to register, manage profiles, and access other verified member businesses through **gated access control**.
- Engineered **dynamic member dashboards** using **Wix Velo (JavaScript)** to manage subscription tiers, listing visibility, and **role-based permissions** via REST APIs.
- Designed and implemented a **scalable data ingestion pipeline** integrating automated CSV workflows with structured CMS collections to streamline business onboarding.
- Developed **subscription-gated access and validation systems** with integrated **analytics tracking** to monitor engagement, subscription activity, and vendor interactions.

Class Representative & Peer Note-Sharing Volunteer

Sep. 2025 – Present

Student Accessibility Services (SAS), York University

Toronto, ON

- Represented students in **MATH 1013**, **MATH 1028**, and **PHYS 1801**, acting as a liaison between students and instructors.
- Organized calculus review sessions supporting academic performance and accessibility standards.

Event Coordinator

Sep. 2025 – Dec. 2025

Iranian Students Association at York University (ISAYU)

Toronto, ON

- Planned and coordinated cultural events, managing logistics, university policy compliance, and safety protocols.
- Collaborated with executive team members, external vendors, and university departments to execute high-attendance events while maintaining budget control and operational efficiency.

PROJECTS

UniSpot | Go, React, PostgreSQL, Supabase, Mapbox, WebSockets

Feb. 2026 – Present

- Architected and deployed a real-time campus intelligence platform for York University students using **React** and **Go (Gin)**.
- Engineered a **JWT** synchronization pipeline using **Gorilla WebSockets** and **PostgreSQL LISTEN/NOTIFY** to propagate INSERT, UPDATE, and DELETE events across all connected clients.
- Implemented advanced geospatial querying with **PostGIS** (**ST_DWithin**, **ST_GeogFromText**) and interactive mapping via **Mapbox GL**.
- Designed an IP-based verification system using **Supabase** with composite unique constraints to prevent vote manipulation.
- Optimized frontend state updates using WebSocket-driven in-place patching to eliminate full re-renders and ensure high-performance UI responsiveness.

Undercut | *Go, FastAPI, Next.js, PostgreSQL, Docker*

Jan. 2026 – Feb. 2026

- Developed a live full-stack platform aggregating thousands of GTA car listings, utilizing a concurrent **Go** scraper to identify underpriced vehicles in real-time.
- Engineered a data-driven **Market Analysis** engine providing a 5-year projected Total Cost of Ownership (TCO) comparison, factoring in depreciation and fuel efficiency.
- Implemented an automated **Deal Grading** system (S-Tier to F-Tier) using quantitative pricing models to benchmark listings against Fair Market Value (FMV).
- Leveraged **Google Gemini 1.5 Flash** for multi-modal analysis of seller descriptions, providing automated red-flag detection and personalized negotiation scripts.
- Designed a responsive frontend with **Next.js 14** and **Tailwind CSS**, featuring interactive market visualizations and a persistent "Garage" for user-saved deals.

HawkEye | *Python, Flask, Google Gemini API, Docker, Tailwind CSS*

Jan. 2026

- Built an AI-powered multi-modal analysis tool that converts raw inventory videos into structured data listings with automated summarization.
- Integrated **Google Gemini 2.5 Flash** to perform advanced visual classification and natural language extraction from video frames.
- Engineered a custom **FFmpeg** pipeline for high-efficiency frame extraction and timestamp analysis to optimize AI processing time.
- Containerized the entire application using **Docker** to ensure environment parity and scalable deployment on Render.

LassondeHub | *Supabase, PostgreSQL, JavaScript, Tailwind CSS, Vercel*

Jan. 2026

- Built a centralized engineering resource platform with secure, role-based access to academic materials.
- Implemented backend infrastructure using **Supabase** and **PostgreSQL**, integrating **Supabase Auth** for secure session management.
- Deployed on **Vercel** and designed a responsive UI with **Tailwind CSS**, supporting dark mode and modern glassmorphism styling.

AlgoDrift | *Next.js, TypeScript, Tailwind CSS, Framer Motion*

Dec. 2025

- Developed an interactive web-based platform for visualizing fundamental data structures and algorithms using **Framer Motion** for fluid state transitions.
- Implemented dynamic, real-time simulations for Linear Search, Binary Search, and Bubble Sort, allowing users to control execution speed and data input.
- Designed a responsive UI with step-by-step execution highlighting to improve pedagogical clarity and user engagement.

TECHNICAL SKILLS**Languages:** JavaScript (ES6+), Python, Go (Golang), C++, HTML5, CSS3, MATLAB**Frontend:** React, Next.js, Tailwind CSS, Mapbox GL, Vite, Framer Motion, Supabase Auth**Backend & Databases:** Node.js, Express, WebSocket, Supabase, PostgreSQL, Firebase**Tools & AI:** Git, GitHub, Docker, Linux, Vercel, Render, OpenCV